

DAWEI(DAVID) XIAO

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Personal Homepage

EDUCATION

Master of Science in Physics and Astronomy | *Theoretical Physics Track*
University of Amsterdam

Sep. 2024 – Aug. 2026
Cum Laude (8.5/10)

Master of Science | *Physics*
King's College London

Sep. 2022 – Sep. 2023
Pass with Distinction (75)

Bachelor of Engineering with Honours | *Materials Science and Engineering*
Queen Mary University of London

Aug. 2018 – Aug. 2022
Upper Second-Class Honours (67.5)

Bachelor of Engineering | *Materials Science and Engineering*
Northwestern Polytechnical University (joint program with QMUL)

Aug. 2018 – Aug. 2022
83.5/100 (Top 26%)

PROJECTS AND RESEARCH

On the phase of the de Sitter Partition Function

Sep. 2025 – Aug. 2026

Supervisor: Dr. Jan Pieter van der Schaar, University of Amsterdam

- The saddle-point approximation to the gravitational partition function evaluated on pure (Euclidean) de Sitter spacetime reproduces the Gibbons–Hawking entropy of the cosmological horizon. Together with the compactness of Euclidean de Sitter space, this result appears to suggest a state-counting interpretation. However, the one-loop correction contains a divergent Gaussian direction, which gives rise to a phase of i after Wick rotation. As a result, the partition function on Euclidean de Sitter space, S^4 acquires a dimension-dependent phase, contradicting the state-counting interpretation.
- We review this long-standing problem and discuss a recent proposal to resolve it by introducing an observer and properly accounting for the Hamiltonian constraint. We argue that the partition function of a charged de Sitter black hole should be understood as coupling two microcanonical ensembles defined with York boundaries. Using a toy model, we compute the net phase of this coupled system, show that it is always real, and identify the corresponding phase transition line. Motivated by recent discussions of the observer dependence of black-hole heat capacity, we further show that the choice of observer also affects the one-loop determinant. Finally, we comment on how the recent observer-based proposal can be recovered from our model.

Application of the Newman-Janis Algorithm to wormhole spacetimes

Feb. 2023 – Aug. 2023

Supervisor: Prof Sarben Sarkar, King's College London

- The Newman-Janis (NJ) Algorithm is an *off-shell* solution-generating algorithm that generates a stationary axisymmetric rotating solution from a static solution. Recently, it was found this algorithm can be generalized to the Taub-NUT metric, the non-zero cosmological constant case, and even the electromagnetic field utilizing the Einstein field equation. We review various extensions and explanations of this algorithm.
- We examine the application of the Newman–Janis algorithm to the Morris–Thorne wormhole and find that it fails to generate a rotating solution; the resulting metric remains static. We provide an explanation for this failure based on the Simpson–Visser substitution.

First Principles Investigations of Symmetric Tilt Grain Boundary in γ -TiAl

Nov. 2021 – Apr. 2022

Supervisor: Prof Hu Rui, NPU and Prof Andy Bushby, QMUL

- Grain boundary (GB) in polycrystalline materials will significantly influence their microstructure evolution and mechanical properties. We constructed the atomic structures of γ/γ tilt GBs in the γ -TiAl phase and analysed their structures and energetics using the first-principles method.
- We systematically analyzed the GB formation energy, excess free volume and interplanar spacing of the GB face of models. We further analyzed the superlattice type of the model and studied the effect of Ta segregation on GB formation energy. Finally, we verified the existence of some low-energy grain boundaries by experiments.

PUBLICATIONS

- Gao, Z., Hu, R., Xiao, D., *et al.* "First-principles investigation on electronic structures and energetic characteristics of γ/γ tilt grain boundaries in γ -TiAl intermetallic," *Intermetallics*, vol. 151, 2022, 107723.

HONORS AND AWARDS

The Scholarship of QMUL Engineering School, NPU

Sep. 2021

Recognition for students in academics at QMES over the academic year

China College Students' 'Internet +' Innovation and Entrepreneurship Competition

Nov. 2020

International Track Silver Award

The Scholarship of QMUL Engineering School, NPU

Sep. 2019

Recognition for students in academics at QMES over the academic year

SKILLS

Languages: Mandarin (Native), English (IELTS 7.0, taken in 2022).

Programming: LaTeX, Mathematica, Material Studio and Maple.

Hobbies: Football, Board games, Synth music and light music.

REFEREES

Dr. Jan Pieter van der Schaar, Institute of Physics, University of Amsterdam

J.P.vanderSchaar@uva.nl

Prof. Sarben Sarkar, Department of Physics, King's College London

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